Local Validation and Workflow Integration of Pathology Foundation Models for Colorectal WSI Classification

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Abstract. Clinical translation of computational pathology models requires both robust validation across institutions and a defined strategy for integrating model outputs into diagnostic workflows. We investigated this problem for three-class colorectal whole-slide image classification into non-neoplastic tissue, low-grade lesions, and high-grade lesions. Five pathology foundation models were evaluated across 12 training and aggregation settings using public development data and an independent local-hospital cohort. Representation shift was measured repeatedly in the original foundation-model embeddings and at successive attention-based multiple-instance learning layers. We then evaluated grade-based and confidence-based case allocation in a reader study with one expert and one novice pathologist with escalation when the novice diagnosis disagreed with the AI prediction. Foundation-model choice was the main determinant of local performance, whereas downstream setting variation had a smaller effect. Run-level shift metrics were weak predictors of performance loss, but foundation-model-level shift rankings corresponded to sensitivity across downstream settings. In the reader study, AI alone achieved an MCC of 0.834 and high-grade sensitivity of 81.1%; the observed human workflow after expert review of escalations achieved 0.808 and 71.6%, respectively. Confidence-based routing captured more AI errors than grade-based routing (72.7% versus 26.3%), without increasing expert workload, whereas displaying AI output did not improve pathologist performance or alter escalation frequency.

Keywords: Computational pathology · Foundation models · Domain shift · AI-assisted triage · Clinical workflow integration

1 Introduction

Colorectal cancer is the third most frequently diagnosed cancer and the second leading cause of cancer death worldwide [1]. Most sporadic cases arise through

adenomatous precursor lesions, for which histological grading, particularly the detection of high-grade dysplasia, informs surveillance and clinical management; high-grade dysplasia is associated with increased risk of subsequent advanced neoplasia [22]. As diagnostic volumes increase, cases are often first encountered by pathologists with variable experience, making efficient allocation to expert review clinically important. Triage systems could therefore support timely reporting while directing specialist attention where it is most needed [15].

Computational pathology can support this by using weakly supervised learning to aggregate patch representations into slide-level predictions [3,14]. Pathology foundation models (FMs) provide general-purpose patch encoders with strong downstream performance across tasks [2,16].

Despite these advances, local translation remains difficult. Models are often trained elsewhere and transferred to a target hospital, where differences in staining, processing, scanners, institutions, and patients can shift FM representations and reduce performance [21,9]. Some studies have proposed compensating for domain shift through feature adaptation or alignment [10,12], whereas others have investigated whether unlabeled feature-space distances can quantify shift and anticipate performance degradation [18]. Finally, deployment requires not only robust model selection, but also a workflow strategy, such as routing cases by predicted risk or model uncertainty, with different implications for workload and safety [6,24].

In this study, we simulate the translation of a colorectal WSI classifier from public development data to a local hospital setting. We evaluate five pathology FMs across multiple downstream settings, measure both performance drop and representation shift from development to local cohorts, and test whether shift metrics explain performance loss or setting sensitivity. We then carry the selected model into a reader study comparing grade-based and confidence-based routing (Fig. 1). Our results show that FM choice, more than downstream setting variation, drives local performance and stability; shift metrics are weak run-level predictors but informative at FM level; and confidence-based routing captures model errors better than grade-based routing.

Our main contributions are:

- a systematic development-cohort and local-hospital evaluation of five pathology FMs across multiple training settings for three-class colorectal WSI classification;
- a repeated, layer-wise analysis of representation shift and its relationship with local-validation performance degradation and model stability; and
- a reader-study comparison of risk- and uncertainty-based case allocation in a human-supervised pathology workflow.

2 Methods

2.1 Datasets

We used the IMP Colorectal Sampling (IMP-CRS) dataset for model development, retaining its labels (847 non-neoplastic, 2,847 low-grade lesions, and 1,639

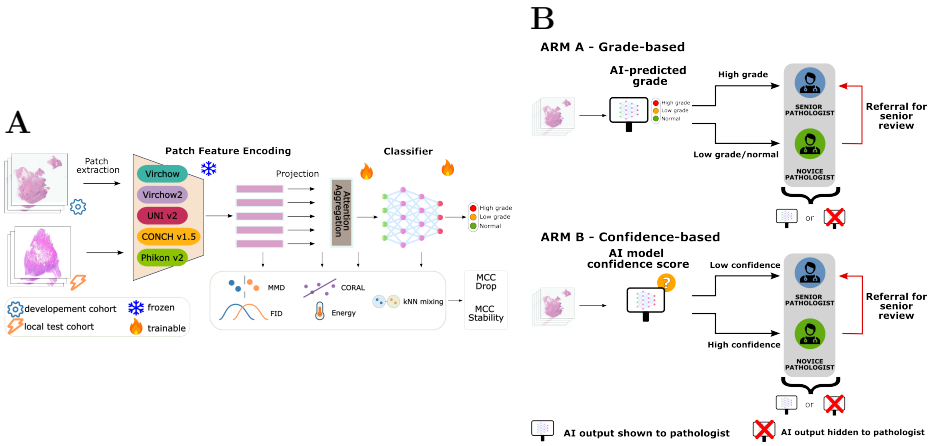


Fig. 1. Overview of the study. **A**, Five pathology foundation models were evaluated using frozen patch encoders and trainable ABMIL classifiers. Performance and representation shift were assessed between the development and local hospital cohorts using shift measures at different layers. **B**, Reader-study design comparing grade-based and confidence-based case allocation to senior and novice pathologists, with referral for senior review upon disagreement with the AI prediction.

high-grade lesions) and original splits [17]. Local validation used 152 colorectal WSIs from Santa Chiara Hospital, Trento: 43 non-neoplastic, 75 low-grade lesions, and 34 high-grade lesions. By IMP-CRS definitions, the high-grade class included high-grade dysplasia, intramucosal carcinoma, and invasive adenocarcinoma. The study was approved by the local ethical committee.

2.2 Model Training

Patch features were extracted with five pathology FMs: UNI v2 [4], CONCH v1.5 [13], Virchow [25], Virchow2 [27], and Phikon v2 [5]. Slide-level classification used ABMIL [11]. We trained 60 models varying sampled patch count, attention architecture, hidden dimension, dropout, learning rate, and weight decay; for the full details refer to the project repository. Models were trained with class-weighted cross-entropy, AdamW, and validation macro-F1 early stopping. WSI pre-processing, feature extraction, and ABMIL implementation used the Trident package [26,23].

2.3 Performance and Representation-Shift Analysis

Performance was measured by multiclass MCC, and performance drop was defined as $\mathrm{MCC}_{\mathrm{development}} - \mathrm{MCC}_{\mathrm{local}}$. Representation shift was quantified with MMD, Fréchet distance, CORAL, energy distance, and 1−k-nearest-neighbour domain mixing ($k = 15$) after standardizing features with development-cohort

statistics [7,8,19,20]. Shift was measured in the original FM mean-patch embedding and, for each ABMIL model, in mean pre-attention, attention-pooled, penultimate, and top-64 attention-weighted patch representations [18]. Repeat-specific MCC drops were computed on the same class-stratified samples of 80 development-test and 80 local-validation slides over 20 repetitions.

Associations with MCC drop were evaluated by Spearman correlation at run and FM levels. Setting sensitivity was defined as the standard deviation of MCC drop across training settings for each FM and repetition; grand-average FM-level analyses used one mean shift–sensitivity pair per FM with exact permutation testing.

2.4 Reader Study

The selected model was evaluated in a reader-study workflow simulation comparing two human–AI case-allocation strategies (Fig. 1B). In Arm A, grade-based triage, cases with predicted high-grade probability higher than 0.20 were assigned to the expert pathologist; all others went to the non-expert. In Arm B, confidence-based triage, uncertain cases were assigned to the expert, where uncertainty was defined as normalized predictive entropy at least 0.30 or prediction margin was less than or equal to 0.20. The routing thresholds were chosen as operational cutoffs to balance workload; sensitivity analyses showed that changes around these values did not substantially alter the main routing outcomes. Each assigned case was read once, with AI predictions shown for half of cases.

One gastrointestinal pathology expert and one pathology trainee participated. Novice cases were sent to expert review when the novice diagnosis disagreed with the AI prediction (escalation), and the expert diagnosis was considered final. The study included 300 cases: 152 local-validation cases, 138 IMP-CRS development-test cases (added to increase sample size and balance the classes between the two arms). We also included 10 sessile serrated lesions (SSLs) from the local hospital, five in each arm, used as an out-of-distribution set (as the model has not seen this class). Readers also rated case difficulty on a scale of 1 to 5 (5 being the most difficult).

3 Results

3.1 Model Validation Results

Across the 60 ABMIL models, development-test MCC was consistently high, with FM means ranging from 0.905 to 0.924 (Table 1). Performance decreased on the local validation cohort for every FM, where mean MCC ranged from 0.724 to 0.762 and variability between runs increased. The distribution of MCC drop remained primarily separated by FM rather than by the tested settings (Fig. 2A). Univariable comparisons of training settings showed no consistent effect across the development and local cohorts (Fig. 2B). Only learning rate showed one nominal association with local-cohort MCC ($p = 0.0373$), which

Table 1. Performance results for different foundation models. Values are mean ± standard deviation across 12 settings per FM. Bold and underlined values denote the best and second-best results, respectively.

FM	Development test cohort MCC	Local validation cohort MCC	MCC drop
Virchow2	**0.924 ± 0.005**	0.748 ± 0.031	0.176 ± 0.031
Virchow	0.913 ± 0.008	0.748 ± 0.035	0.166 ± 0.036
CONCH v1.5	0.905 ± 0.007	**0.762 ± 0.035**	**0.143 ± 0.032**
UNI v2	<u>0.920 ± 0.011</u>	0.724 ± 0.065	0.195 ± 0.056
Phikon v2	0.913 ± 0.006	<u>0.750 ± 0.080</u>	<u>0.163 ± 0.082</u>

was not mirrored in development performance. Notably, CONCH v1.5 had the lowest mean MCC on the development test cohort, but achieved the highest mean MCC on the local validation cohort, the smallest performance drop, and the lowest instability.

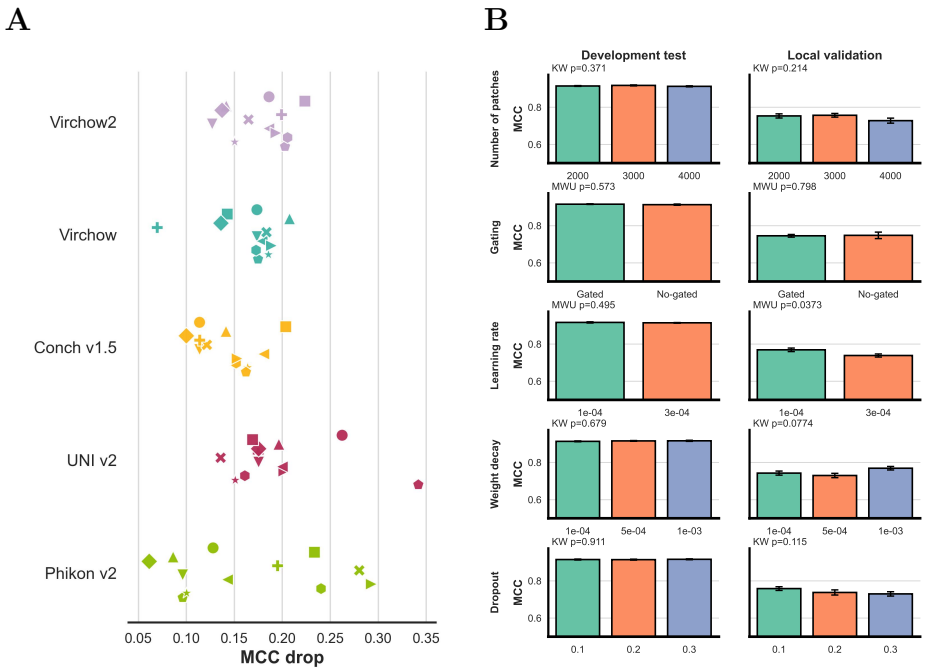


Fig. 2. Influence of FM choice and downstream training and aggregation settings on validation performance. **A**, Development-to-local MCC drop for all 60 trained models. Color denotes the FM; marker shape distinguishes the 12 downstream settings. **B**, Mean ± SEM MCC in the development test cohort (left) and local validation cohort (right), stratified by individual settings. Reported p values are from two-sided Mann–Whitney U (MWU) or Kruskal–Wallis (KW) tests, as appropriate.

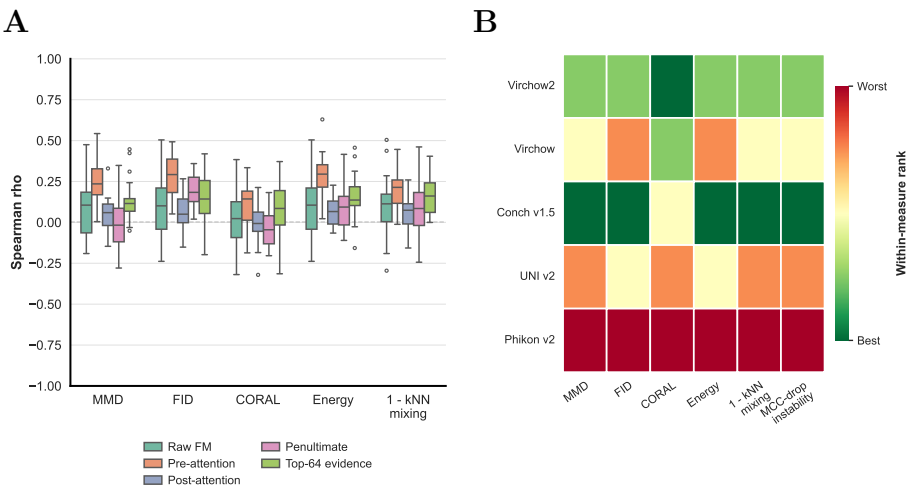


Fig. 3. Representation shift and robustness. **A**, Run-level Spearman correlations between shift metrics and MCC drop across 20 matched repetitions. **B**, FM-level ranks of raw-feature shift and MCC-drop instability averaged across repetitions. Green indicates lower values and red higher values; instability is the standard deviation of MCC drop across the 12 settings within each FM.

For each repetition, each shift measurement was correlated with the repetition-matched MCC drop across all 60 ABMIL runs. Figure 3A shows the resulting distribution of 20 Spearman correlations. Associations with MCC drop were generally weak. The largest and most consistent correlations occurred in the mean pre-attention representation: median ρ was 0.24 for MMD, 0.29 for Fréchet distance, 0.14 for CORAL, 0.29 for energy distance, and 0.21 for 1−kNN mixing. Raw FM features had lower median correlations (ρ = 0.02–0.11), while post-attention and penultimate representations were not consistently more informative. Selecting the 64 patches with the greatest positive attention evidence as proposed in [18] produced modest correlations, but did not outperform the pre-attention mean.

We next asked whether representation shift reflected *FM-level robustness* rather than the MCC drop of individual runs. For each repetition and FM, we calculated setting sensitivity as the standard deviation of MCC drop across the 12 settings. We then averaged setting sensitivity and raw-feature shift metrics across the 20 repetitions to obtain one FM-level value for each measure (Fig. 3B). CONCH showed the most favorable overall profile, with the lowest grand-average shift for most metrics and the lowest setting sensitivity, whereas Phikon v2 showed the largest shift and greatest sensitivity. Across FMs, larger raw-feature shifts generally aligned with greater setting sensitivity.

3.2 Reader Study Results

The highest performing CONCH model was carried forward to the reader study. Across both arms, the AI alone achieved an MCC of 0.834 and high-grade (HG) sensitivity of 60/74 (81.1%), compared with an MCC of 0.585 and HG sensitivity of 47/74 (63.5%) for the initially assigned pathologists. Using the recorded expert diagnosis for every escalated case increased workflow performance to an MCC of 0.808 and HG sensitivity of 53/74 (71.6%).

Table 2. Reader-study performance, AI visibility, and workload after excluding SSL cases. HG values are sensitivity. "Initial" combines the readers according to the initial allocation; "final" replaces escalated novice diagnoses with the recorded expert diagnosis. Error capture is the proportion of AI errors assigned directly to the expert.

Arm	Evaluation	n	Overall		AI shown		AI hidden	
			MCC	HG	MCC	HG	MCC	HG
A	AI	145	0.790	75.7%	0.827	80.0%	0.753	70.6%
	Expert	38	0.751	89.3%	0.839	93.8%	0.675	83.3%
	Novice	107	0.428	22.2%	0.374	0.0%	0.491	40.0%
	Initial workflow	145	0.572	73.0%	0.569	75.0%	0.581	70.6%
	Final workflow	145	0.781	67.6%	0.830	75.0%	0.730	58.8%
B	AI	145	0.879	86.5%	0.897	94.7%	0.865	77.8%
	Expert	44	0.784	55.6%	0.797	80.0%	0.793	25.0%
	Novice	101	0.503	53.6%	0.477	42.9%	0.534	64.3%
	Initial workflow	145	0.604	54.1%	0.600	52.6%	0.610	55.6%
	Final workflow	145	0.835	75.7%	0.850	78.9%	0.820	72.2%

Routing and expert workload

Arm	Initial expert	Novice	Escalations shown	Escalations hidden	Total expert load	Error capture
A	38	107	14/54 (25.9%)	18/53 (34.0%)	70/145 (48.3%)	5/19 (26.3%)
B	44	101	17/51 (33.3%)	13/50 (26.0%)	74/145 (51.0%)	8/11 (72.7%)

OOD sessile serrated lesion recognition

Arm	Assigned reader	OOD cases	Initially recognized	Escalated	Recognized after workflow
A	Novice	5	0/5 (0%)	2/5 (40%)	0/5 (0%)
B	Expert	5	5/5 (100%)	0/5 (0%)	5/5 (100%)

The two strategies produced nearly identical escalation rates (Table 2) and similar total expert workloads: 70/145 cases (48.3%) in Arm A and 74/145 (51.0%) in Arm B. Their principal difference was therefore the composition of the expert queue. Grade-based routing sent 28/37 HG cases directly to the expert, compared with 9/37 under confidence routing. Conversely, confidence routing captured a larger proportion of AI errors in the initial expert queue (72.7% versus 26.3%; Fisher exact $p = 0.023$). Arm B had the higher final MCC, although the bootstrap 95% CI for the between-arm difference included zero ($\Delta = 0.055$, 95% CI -0.061 to 0.172). Final HG sensitivity also did not differ significantly between arms ($p = 0.607$).

Displaying the AI output did not improve initial pathologist MCC, HG sensitivity, or escalation frequency (Table 2). Overall escalation occurred in 29.5% of shown and 30.1% of hidden novice cases ($p = 1.00$), indicating that visibility did not suppress disagreement or referral. Reader–AI agreement was also unchanged (74.8% versus 74.5%). When the AI was wrong, readers recovered 5/13 shown and 6/17 hidden cases; when it was correct, readers changed the prediction to an incorrect diagnosis in 32/134 and 28/124 cases, respectively. Although final

workflow MCC was numerically higher for shown than hidden cases (0.839 versus 0.775), the bootstrap confidence interval included no difference and the AI itself also performed better on the shown subset.

Human and AI performance depended on when expert review occurred. Among the 82 cases assigned directly to the expert, expert MCC was 0.816 versus 0.755 for the AI: the expert corrected eight AI errors, whereas the AI was correct in five cases missed by the expert. The AI nevertheless had higher HG sensitivity (89.2% versus 81.1%). In contrast, on the 62 escalated novice cases, the AI outperformed the expert adjudication (MCC 0.808 versus 0.523; exact McNemar $p = 0.021$) and had higher HG sensitivity (75.0% versus 50.0%). Thus, escalation corrected many novice errors, but expert review did not consistently improve upon the AI prediction itself.

Perceived difficulty supported the confidence-routing rationale. Incorrect diagnoses were rated more difficult than correct diagnoses overall (median 3 [IQR 2–4] versus 2 [IQR 1–3], $p < 10^{-7}$). Escalated cases were also more difficult than non-escalated novice cases (median 3 versus 2, $p = 0.0025$). Finally, the SSLs were evaluated as a separate out-of-distribution endpoint: all five assigned directly to the Arm B expert were recognized as SSLs, whereas none of the five assigned initially to the Arm A novice were recognized.

4 Conclusion

We evaluated the transfer and clinical integration of a colorectal WSI classifier developed using an external public cohort and deployed on local hospital data. Across 60 trained ABMIL models, FM selection had a greater influence on local performance and robustness than variation in downstream training settings. Unsupervised representation-shift measures did not reliably predict the performance loss of an individual trained model; however, when aggregated by FM, they identified broad differences in sensitivity to downstream settings. Such measurements may therefore support FM-level robustness assessment.

The reader study further showed that implementation strategy was more consequential than AI visibility alone. Grade-based routing prioritized HG cases, whereas confidence-based routing captured substantially more AI errors with a similar expert workload. Displaying the AI output did not improve MCC, HG sensitivity, or escalation frequency. Expert adjudication corrected many novice errors, but AI remained superior on the escalated case subset. Finally, recognition of rare cases depended strongly on reader allocation. Because this pilot involved one expert and one novice pathologist, the findings are preliminary and motivate larger multi-reader, multi-institution evaluations of workflow-specific routing strategies. Overall, safe clinical integration requires explicit local validation, routing aligned with the clinical objective, and empirical evaluation of the entire human–AI pathway, including the consequences of disagreement and escalation.

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Disclosure of Interests. The authors declare no competing interests.

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